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# Online Quiz System

## TECHNICAL REPORT



### SUBMITTED BY

Zainab (2024-ag-9715)

### ADVISED BY

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**A TECHNICAL REPORT SUBMITTED IN PARTIAL FULFILLMENT OF  
REQUIREMENT FOR THE DEGREE OF**

*BACHELOR OF SCIENCE*

*IN*

*COMPUTER SCIENCE*

**DEPARTMENT OF COMPUTER SCIENCE**

**FACULTY OF SCIENCES**

**UNIVERSITY OF AGRICULTURE FAISALABAD**

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## DECLARATION

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I hereby declare that the contents of the report "Online Quiz System" is a project of my own and no part has been copied from any published source (except the references). I further declare that this work has not been submitted for award of any other diploma/degree. The university may act if the information provided is found false at any stage. In case of any default the scholar will proceed against as per UAF policy.

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Zainab

2024-ag-9715

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## CERTIFICATE

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To,

The Controller of Examinations,  
University of Agriculture,  
Faisalabad.

The supervisory committee certifies that Zainab bearing AG# 2024-ag-9715 has successfully completed her project "Online Quiz System" in partial fulfillment of requirement for the degree of B.Sc. Computer Science under our guidance and supervision.

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Advisor Name:

Sir Qamar Nawaz

Department of Computer Science

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Dr. Muhammad Ahsan Latif

Chairman,

Department of Computer Science

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## ACKNOWLEDGEMENT

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I thank all who contributed in one way or another to the completion of this report. I am so grateful to the Department of Computer Science, University of Agriculture Faisalabad, for making it possible for me to pursue my studies here. My special and heartfelt thanks go to my advisor, Sir Qamar, who encouraged and directed me throughout every phase of this project. His guidance and constructive challenges brought this work to completion. It is with his supervision that this work came into existence. For any shortcomings, I take full responsibility.

I am also thankful to my fellow students whose productive critiques and collaborative spirit provided new ideas and perspectives for the work. Furthermore, I thank my family who continuously encouraged me and prayed for me throughout the entire period of my project. May the Almighty God richly bless all of you.

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## ABSTRACT

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The Online Quiz System is a desktop-based application developed using C# and .NET Framework to digitize and streamline the process of creating, managing, and evaluating quizzes within an academic environment. The system uses JSON (JavaScript Object Notation) files as the primary data storage format, implemented through C# file handling and Newtonsoft.Json serialization. This eliminates the need for a traditional relational database and makes the system lightweight, portable, and easy to deploy without requiring any database server installation.

The system supports two distinct user roles: Teacher (Admin) and Student. Authentication is handled through a Login Form where user credentials are validated against a JSON file. After successful login, the system redirects users to their respective dashboards based on their roles.

The Teacher Dashboard provides core functionalities including creating quizzes through the Create Quiz Form, viewing all student results through the View Result Form, and logging out of the system. The Create Quiz Form allows teachers to define quiz titles, questions, and multiple-choice options, which are serialized into JSON format and stored in local files.

The Student Dashboard allows students to attempt quizzes using the Quiz Picker Form, which dynamically loads available quizzes from JSON files and randomly selects one at runtime. After completion, the student's responses are evaluated, and results are stored in JSON format. Students can also view their individual results through the shared View Result Form, which filters data based on the logged-in user.

The system is developed using C# Windows Forms in .NET Framework, following a modular design approach. JSON-based file storage is used for managing users, quizzes, and results. The system is efficient, user-friendly, and suitable for small-scale academic environments that require a simple, offline, and server-free quiz management solution.

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## CHAPTER 1 - INTRODUCTION

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### 1.1 Background

The growing integration of technology in education has transformed the way academic assessments are conducted. Traditional paper-based quizzes, while familiar, carry numerous inefficiencies including high paper consumption, manual grading errors, delayed feedback, and considerable overhead time for educators. In response to these challenges, educational institutions are increasingly turning to software-based quiz systems that automate the entire assessment lifecycle.

The Online Quiz System was initiated to address the specific needs of a classroom or small academic department that requires a simple, efficient, and self-contained tool for quiz management. Unlike complex web-based Learning Management Systems (LMS) that require server infrastructure, internet connectivity, and administrative overhead, this system is designed as a standalone desktop application that operates entirely on local storage using JSON as its data format. This design decision makes it accessible even in resource-constrained environments.

The system was conceived with two primary users in mind: the Teacher, who needs to create and manage quizzes efficiently, and the student, who needs to attempt quizzes and receive immediate feedback. The use of JSON packages eliminates the dependency on a separate database management system, as all quiz data, user credentials, and result records are stored in structured JSON files. This approach is particularly advantageous for academic projects where simplicity, portability, and ease of deployment are priorities. The system's design ensures that any instructor with basic computer literacy can adopt it without specialized training. [1]

### 1.2 Description

The Online Quiz System is a desktop application developed using C# and .NET Framework with Windows Forms. The system consists of multiple forms that handle authentication, quiz management, quiz attempts, and result display.

The Login Form is the entry point of the system. It contains username and password fields and validates user credentials using a JSON file. Based on the role (Teacher or Student), users are redirected to their respective dashboards.

The Teacher Dashboard provides access to Create Quiz, View Result, and Logout functions. The Create Quiz Form allows teachers to enter quiz details including questions, options, and correct answers. This data is stored in JSON format on the local file system.

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The Student Dashboard allows students to start quizzes, view results, and logout. When a student starts a quiz, a Quiz Picker Form loads available quizzes from JSON files and randomly selects one for attempt. After submission, results are calculated and stored in JSON format.

The system uses C# file handling techniques with JSON serialization and deserialization for all data operations.

### 1.3 Problem Statement

Academic institutions conducting regular quizzes face operational inefficiencies caused by manual, paper-based assessment processes. Teachers invest significant non-instructional time in printing quiz papers, manually distributing them, collecting responses, and grading each paper individually. This process is prone to human error in grading and introduces significant delays in delivering feedback to students, reducing the educational value of the assessment.

Furthermore, storing and retrieving historical quiz records for multiple students is cumbersome with paper-based systems. There is a clear need for a lightweight, offline-capable, easy-to-deploy quiz management system that automates quiz creation, distribution, and result recording, while remaining accessible without requiring internet connectivity or complex server setup. The Online Quiz System, using JSON-based data storage, directly addresses this problem. [3]

### 1.4 Scope

The scope of the Online Quiz System covers the following functional boundaries:

- User authentication for two roles: Teacher (admin) and Student, validated against a JSON credential store.
- Teacher-side operations: Creating quizzes via the Create Quiz Form and storing them as JSON files; viewing all student results via the View Result Form.
- Student-side operations: Starting a randomly assigned quiz via the Quiz Picker Form (runtime only); viewing personal results via the View Result Form.
- A shared View Result Form that displays role-appropriate data — all results for teachers, personal results for students.
- Quiz Picker Form that only appears at runtime when the student initiates a quiz, randomly selecting from available JSON quiz files.
- All data (users, quizzes, results) is stored and managed exclusively through JSON files — no external database required.
- The system is scoped for desktop use only and does not include web browser or mobile compatibility.
- Teacher-side operations now also include: Manage Quizzes

- 
- Form for renaming and deleting existing quizzes (full CRUD).
  - -Student-side operations now also include: Real-time search
  - filtering in Quiz Picker Form to find quizzes by title.

## **1.5 Objectives**

Goal: To develop a fully functional, JSON-powered desktop-based Online Quiz System that automates quiz administration for academic environments.

- Objective 1: Implement a secure Login Form that authenticates admin and student users using JSON-stored credentials and routes them to the correct dashboard.
- Objective 2: Develop a Teacher Dashboard with Create Quiz, View Result, and Logout buttons.
- Objective 3: Build a Create Quiz Form that allows teachers to define quizzes and persist them as JSON files.
- Objective 4: Develop a Student Dashboard with Start Quiz, View Result, and Logout buttons.
- Objective 5: Implement a Quiz Picker Form that appears only at runtime, randomly selects a quiz from JSON storage, and presents it to the student for completion.
- Objective 6: Create a shared View Result Form that reads result JSON files and displays data filtered by the logged-in user's role.
- Objective 7: Ensure the system is portable, lightweight, and operable without an internet connection or database server.

## **1.6 Feasibility**

A feasibility study was conducted to verify the practicality of the Online Quiz System before development commenced.

### **1.6.1 Technical Feasibility**

The development team has strong knowledge of C# programming and .NET Framework. The system is developed using Windows Forms with JSON file handling using Newtonsoft.Json. No external database server is required.

### **1.6.2 Schedule Feasibility**

The system is modular, comprising six independently developable forms. Development can proceed in parallel sprints under the Agile model, making it feasible to complete within a standard academic semester.

### **1.6.3 Economic Feasibility**

The project requires no commercial licenses. JavaScript is freely available, and JSON requires no additional libraries. The economic cost is negligible, making the project fully economically feasible.



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### 1.6.4 Cultural Feasibility

The system supports standard academic quiz practices and introduces no cultural barriers. Its interface is simple and language-neutral, making it culturally appropriate for diverse academic settings.

### 1.6.5 Legal/Ethical Feasibility

All user data (credentials and results) is stored locally in JSON files. No data is transmitted externally, ensuring user privacy. The system complies with standard academic integrity requirements as quiz randomization prevents answer sharing. It is legally and ethically feasible.

### 1.6.6 Resource Feasibility

The system runs on any standard personal computer with a JavaScript runtime. No dedicated server, database engine, or internet connection is required. Resource requirements are minimal and fully available.

### 1.6.7 Operational Feasibility

The application's clean two-role interface requires minimal training. Teachers and students can begin using the system with minimal onboarding. JSON-based architecture also makes troubleshooting and data inspection straightforward, supporting long-term operational feasibility.

## 1.7 Requirements

### 1.7.1 Functional Requirements

#### FR01: User Login

|         |                                                                                                            |
|---------|------------------------------------------------------------------------------------------------------------|
| FR01-01 | System shall display a Login Form with Username and Password fields and a Login button.                    |
| FR01-02 | System shall read user credentials from a JSON file and validate the entered username and password.        |
| FR01-03 | System shall navigate admin/teacher users to the Teacher Dashboard upon successful login.                  |
| FR01-04 | System shall navigate student users to the Student Dashboard upon successful login.                        |
| FR01-05 | System shall display an error message if credentials do not match any record in the JSON credential store. |

#### FR02: Teacher Dashboard

|         |                                                                                      |
|---------|--------------------------------------------------------------------------------------|
| FR02-01 | Teacher Dashboard shall display three buttons: Create Quiz, View Result, and Logout. |
| FR02-02 | Clicking Create Quiz shall open the Create Quiz Form.                                |
| FR02-03 | Clicking View Result shall open the View Result Form showing all students' results.  |
| FR02-04 | Clicking Logout shall end the session and return to the Login Form.                  |

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### FR03: Create Quiz Form

|         |                                                                                                            |
|---------|------------------------------------------------------------------------------------------------------------|
| FR03-01 | System shall display the Create Quiz Form when the teacher clicks the Create Quiz button.                  |
| FR03-02 | Teacher shall be able to enter a quiz title, questions, four options per question, and the correct answer. |
| FR03-03 | System shall serialize all quiz data into a JSON file and save it to local storage upon submission.        |
| FR03-04 | System shall confirm successful save and return teacher to the Teacher Dashboard.                          |

### FR04: Student Dashboard

|         |                                                                                                    |
|---------|----------------------------------------------------------------------------------------------------|
| FR04-01 | Student Dashboard shall display three buttons: Start Quiz, View Result, and Logout.                |
| FR04-02 | Clicking Start Quiz shall open the Quiz Picker Form at runtime.                                    |
| FR04-03 | Clicking View Result shall open the View Result Form showing only the logged-in student's results. |
| FR04-04 | Clicking Logout shall end the session and return to the Login Form.                                |

### FR05: Quiz Picker Form

|         |                                                                                                               |
|---------|---------------------------------------------------------------------------------------------------------------|
| FR05-01 | Quiz Picker Form shall appear only at runtime when the student clicks Start Quiz.                             |
| FR05-02 | System shall read all available quiz JSON files and randomly select one for the student.                      |
| FR05-03 | Selected quiz questions and options shall be displayed to the student for answering.                          |
| FR05-04 | Upon submission, system shall calculate the score, write a result JSON file, and return student to dashboard. |

### FR06: View Result Form

|         |                                                                                                                    |
|---------|--------------------------------------------------------------------------------------------------------------------|
| FR06-01 | System shall open the View Result Form when the View Result button is clicked from either dashboard.               |
| FR06-02 | For Teacher: system shall read all result JSON files and display each student's name, quiz title, score, and date. |
| FR06-03 | For Student: system shall read only the result JSON files belonging to the logged-in student.                      |

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### **1.7.2 Non-Functional Requirements**

#### **NFR01:**

System shall always remain available to its users without requiring internet connectivity.

#### **NFR02:**

System shall support two user roles: admin (Teacher) and student, with role-based access control.

#### **NFR03:**

System shall respond to all user interactions (button clicks, form submissions) within 2 seconds.

#### **NFR04:**

System shall ensure data integrity of all JSON files — no partial writes or corrupt records.

#### **NFR05:**

System shall provide clearly labeled buttons and tooltips on all interactive elements.

#### **NFR06:**

System shall prevent unauthorized access between sessions by clearing session state on logout.

#### **NFR07:**

JSON data files shall be structured in a consistent, human-readable scheme to facilitate maintenance.

### **1.7.3 Hardware Requirements**

Processor: Intel Pentium Core i3 or equivalent

RAM: 2 GB or more

Hard Disk: 200 MB free disk space (for application and JSON data files)

Display: 1024 x 768 resolution or higher

### **1.7.4 Software Requirements**

Operating System: Windows 7 / 8 / 10 / 11

Framework: .NET 10.0

Programming Language: C#

IDE: Visual Studio Community 2022

Libraries: Newtonsoft.Json (JSON.NET)

Data Format: JSON (File-based storage using System.IO)

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## 1.8 Stakeholders

The key stakeholders of the Online Quiz System are identified as follows:

- Teacher / Admin: Creates quizzes, monitors all student results, manages logout. Primary driver of quiz content.
- Student: Attempts randomly assigned quizzes, views personal results. Primary end-user of assessment functionality.
- Department Management: Benefits from improved efficiency and digital record-keeping enabled by the system.
- System Developer (Zainab): Responsible for design, implementation, testing, and maintenance of the application.

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## CHAPTER 2 – MATERIALS & METHODS

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### 2.1 Process Model

The Agile development model was selected for the Online Quiz System. Agile is a Rapid Application Development (RAD) methodology based on iterative and incremental delivery. In the Agile model, components are developed in parallel as mini-projects, time-boxed into sprints, and assembled into a growing prototype that is reviewed at each stage.

Agile was chosen because the Online Quiz System has clearly defined modular components — each form representing a distinct deliverable. Sprint 1 covered the Login Form and credential JSON structure. Sprint 2 addressed the Teacher Dashboard and Create Quiz Form with quiz JSON serialization. Sprint 3 implemented the Student Dashboard and Quiz Picker Form with random JSON quiz selection. Sprint 4 completed the shared View Result Form and result JSON reading. This iterative approach enabled continuous testing and refinement at each sprint boundary. [2]

### 2.2 Tools & Technologies

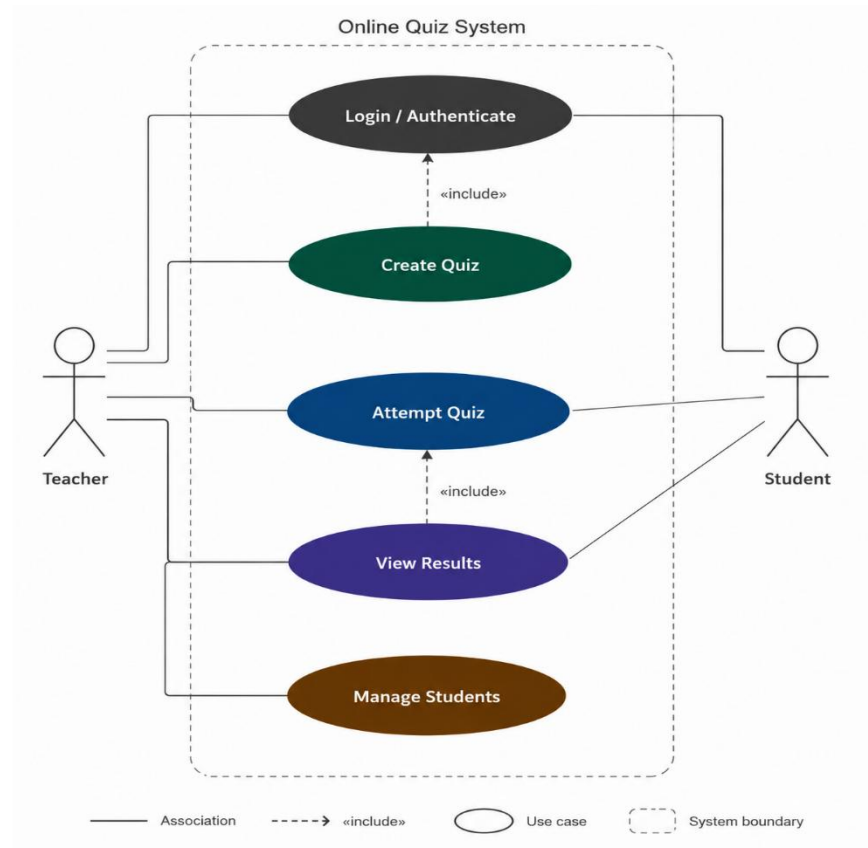
- C# (C-Sharp): Primary programming language used for implementing all application logic.
- .NET Framework: Provides the runtime environment and Windows Forms library for desktop application development.
- Windows Forms (WinForms): Used for designing graphical user interface forms such as Login, Dashboard, Quiz, and Result forms.
- JSON (JavaScript Object Notation): Used as a lightweight data storage format for users, quizzes, and results.
- System.IO Namespace: Used for file handling operations such as reading and writing JSON files.
- Newtonsoft.Json (JSON.NET): Used for serialization and deserialization of objects to and from JSON format.
- Visual Studio Community 2022: Integrated Development Environment (IDE) used for development, debugging, and testing.
- Git: Version control system used for tracking project changes.
- Draw.io: Used for designing UML diagrams, DFDs, and architecture diagrams.

### 2.3 Design

The design section presents all UML and structural diagrams that define the behavior, data flow, and architecture of the Online Quiz System. All diagrams reflect the JSON-based data model adopted for the project. All diagrams are designed using Draw.io to ensure clear visualization of system architecture, data flow, and object relationships.

### 2.3.1 Use Case Diagrams

The Use Case Diagram identifies two actors — Teacher and Student — and five primary use cases that encompass all system interactions.



#### Actors:

- Teacher (Admin): Interacts with Login, Create Quiz, View All Results, and Logout use cases.
- Student: Interacts with Login, Start Quiz (includes Quiz Picker), View Own Results, and Logout use cases.

#### Use Cases:

- UC-01: Login — Both actors authenticate through the Login Form against JSON credential data.
- UC-02: Create Quiz — Teacher opens Create Quiz Form, inputs quiz data, system saves as JSON file.
- UC-03: Start Quiz (includes Quiz Picker) — Student clicks Start Quiz; Quiz Picker Form randomly selects a JSON quiz at runtime.
- UC-04: View Result — Both actors access the shared View Result Form; content filtered by role.
- UC-05: Logout — Both actors terminate their session and return to the Login Form.

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### 2.3.2 Usage Scenarios

#### Usage Scenario — UC-01: Login

|                             |                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Title</b>       | Login                                                                                                                                                                                                                                                                                                                                                         |
| <b>Use Case Id</b>          | UC-01                                                                                                                                                                                                                                                                                                                                                         |
| <b>Requirement Id</b>       | FR01                                                                                                                                                                                                                                                                                                                                                          |
| <b>Description</b>          | A user (Teacher or Student) launches the application and authenticates using the Login Form. The system validates credentials against a JSON file and routes the user to the appropriate dashboard.                                                                                                                                                           |
| <b>Pre Conditions</b>       | Application is running. Login Form is displayed. users.json credential file exists.                                                                                                                                                                                                                                                                           |
| <b>Task Sequence</b>        | 1. User enters username (admin or student ID) and numerical password. 2. User clicks Login button. 3. System reads users.json and checks for a matching username-password pair. 4. If admin: system navigates to Teacher Dashboard. 5. If student: system navigates to Student Dashboard. 6. If no match: system displays an error message on the Login Form. |
| <b>Post Conditions</b>      | Authenticated user is on their correct dashboard. Session state records the logged-in username and role.                                                                                                                                                                                                                                                      |
| <b>Unresolved Issues</b>    | None                                                                                                                                                                                                                                                                                                                                                          |
| <b>Authority</b>            | System (Authentication Controller)                                                                                                                                                                                                                                                                                                                            |
| <b>Modification History</b> | 1.0 — Author: Zainab — Initial version                                                                                                                                                                                                                                                                                                                        |

#### Usage Scenario — UC-02: Create Quiz

|                          |                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Title</b>    | Create Quiz                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Use Case Id</b>       | UC-02                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Requirement Id</b>    | FR02, FR03                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Description</b>       | The logged-in Teacher opens the Create Quiz Form from the Teacher Dashboard, enters quiz content, and saves it as a JSON file.                                                                                                                                                                                                                                                                    |
| <b>Pre Conditions</b>    | Teacher is authenticated and on the Teacher Dashboard.                                                                                                                                                                                                                                                                                                                                            |
| <b>Task Sequence</b>     | 1. Teacher clicks the Create Quiz button. 2. System opens the Create Quiz Form. 3. Teacher enters quiz title, adds questions with four options each, and marks the correct answer. 4. Teacher submits the form. 5. System serializes the quiz object into a uniquely named JSON file (e.g., quiz_001.json) and saves it. 6. System confirms success and returns Teacher to the Teacher Dashboard. |
| <b>Post Conditions</b>   | A new quiz JSON file is saved and becomes available for random selection by the Quiz Picker Form.                                                                                                                                                                                                                                                                                                 |
| <b>Unresolved Issues</b> | None                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Authority</b>         | Teacher                                                                                                                                                                                                                                                                                                                                                                                           |

|                             |                                        |
|-----------------------------|----------------------------------------|
| <b>Modification History</b> | 1.0 — Author: Zainab — Initial version |
|-----------------------------|----------------------------------------|

### Usage Scenario — UC-03: Start Quiz (Quiz Picker)

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Title</b>       | Start Quiz                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Use Case Id</b>          | UC-03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Requirement Id</b>       | FR04, FR05                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Description</b>          | The logged-in Student initiates a quiz attempt. The Quiz Picker Form appears at runtime, randomly selects a quiz from JSON storage, and presents it to the student.                                                                                                                                                                                                                                                                                                                  |
| <b>Pre Conditions</b>       | Student is authenticated and on the Student Dashboard. At least one quiz JSON file exists in the quizzes directory.                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Task Sequence</b>        | 1. Student clicks the Start Quiz button. 2. System opens the Quiz Picker Form (visible only at runtime). 3. Quiz Picker reads all quiz JSON files from the local quizzes directory. 4. Quiz Picker randomly selects one quiz and displays its title and questions with options. 5. Student answers all questions and clicks Submit. 6. System calculates the score, writes a result JSON file (e.g., result_student1_001.json), and navigates student back to the Student Dashboard. |
| <b>Post Conditions</b>      | A result JSON file is saved. Student can view results via View Result Form.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Unresolved Issues</b>    | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Authority</b>            | Student                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Modification History</b> | 1.0 — Author: Zainab — Initial version                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### Usage Scenario — UC-04: View Result

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Title</b>    | View Result                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Use Case Id</b>       | UC-04                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Requirement Id</b>    | FR06                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Description</b>       | Both Teacher and Student access the shared View Result Form. The form reads result JSON files and displays data filtered by role.                                                                                                                                                                                                                                                                                   |
| <b>Pre Conditions</b>    | User is authenticated. Result JSON files exist in the results directory.                                                                                                                                                                                                                                                                                                                                            |
| <b>Task Sequence</b>     | 1. User clicks the View Result button on their dashboard. 2. System opens the View Result Form. 3. If Teacher: system reads all result JSON files and displays a full table (Student Name, Quiz Title, Score, Date). 4. If Student: system reads only result JSON files whose studentId matches the logged-in student and displays them. 5. User reviews results. 6. User clicks Back to return to their dashboard. |
| <b>Post Conditions</b>   | No data is modified. User returns to their dashboard.                                                                                                                                                                                                                                                                                                                                                               |
| <b>Unresolved Issues</b> | None                                                                                                                                                                                                                                                                                                                                                                                                                |



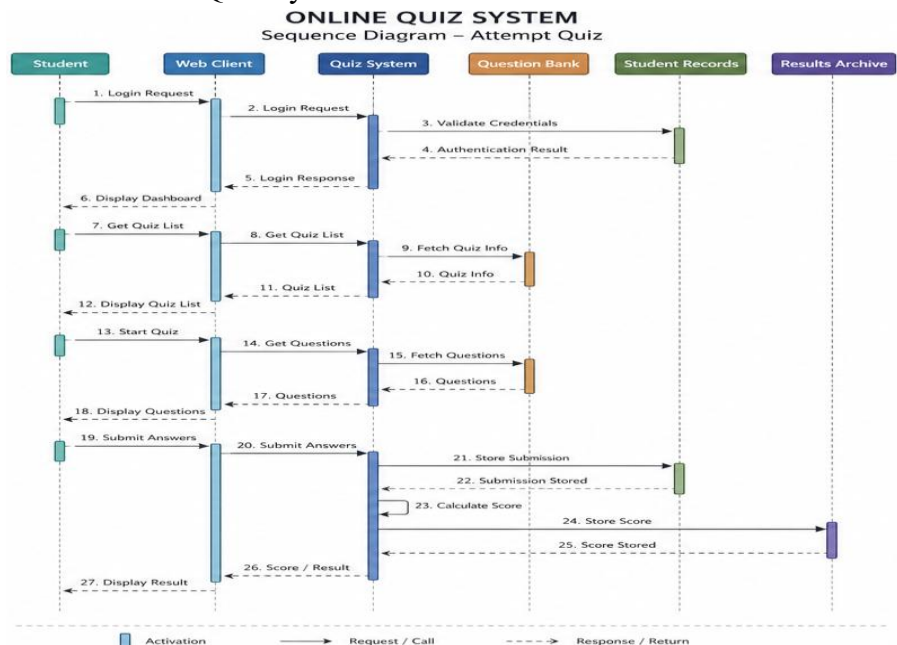
|                             |                                        |
|-----------------------------|----------------------------------------|
| <b>Authority</b>            | Teacher, Student                       |
| <b>Modification History</b> | 1.0 — Author: Zainab — Initial version |

## Usage Scenario — UC-05: Logout

|                             |                                                                                                                                               |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Title</b>       | Logout                                                                                                                                        |
| <b>Use Case Id</b>          | UC-05                                                                                                                                         |
| <b>Requirement Id</b>       | NFR06                                                                                                                                         |
| <b>Description</b>          | Either actor ends their session by clicking Logout on their dashboard.                                                                        |
| <b>Pre Conditions</b>       | User is authenticated and, on either Teacher, or Student Dashboard.                                                                           |
| <b>Task Sequence</b>        | 1. User clicks the Logout button. 2. System clears the session state (current username and role). 3. System navigates back to the Login Form. |
| <b>Post Conditions</b>      | Session is terminated. Login Form is displayed.                                                                                               |
| <b>Unresolved Issues</b>    | None                                                                                                                                          |
| <b>Authority</b>            | Teacher, Student                                                                                                                              |
| <b>Modification History</b> | 1.0 — Author: Zainab — Initial version                                                                                                        |

### 2.3.3 Sequence Diagram

Sequence diagrams depict the order of interactions between objects. Three key sequences are described for the Online Quiz System.



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### **Login Sequence:**

Actor sends credentials to the Login Form UI. The UI invokes the AuthController, which calls the JSON File Handler to read users.json. The file handler returns the parsed user array. AuthController compares credentials and returns role. UI navigates to the appropriate dashboard.

### **Create Quiz Sequence:**

Teacher clicks Create Quiz on Teacher Dashboard UI. UI opens the Create Quiz Form. Teacher inputs quiz data and clicks Save. The form invokes the QuizController. QuizController serializes the quiz object to JSON via the File Handler, which writes quiz\_XXX.json to disk. QuizController returns success. UI confirms and navigates back to Teacher Dashboard.

### **Start Quiz (Quiz Picker) Sequence:**

Student clicks Start Quiz on Student Dashboard UI. UI instantiates the Quiz Picker Form. Quiz Picker invokes the QuizLoader, which reads all .json files from the quizzes directory. QuizLoader randomly selects one quiz and returns it. Quiz Picker renders the quiz to the student. Student submits answers. ResultController calculates score and writes result\_studentID\_XXX.json via the File Handler.

## **2.3.4 Class Diagram**

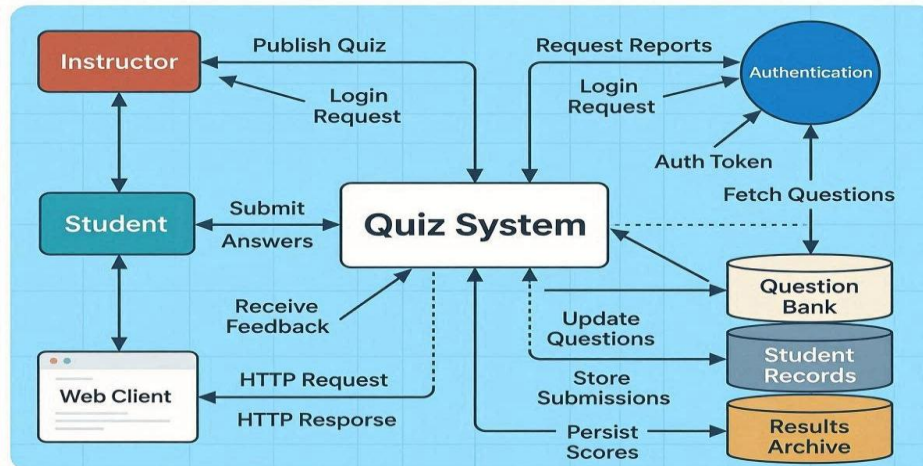
The class structure of the Online Quiz System is organized as follows:

- User {userId, username, password, role} — login(), logout()
- Teacher extends User — openCreateQuizForm(), viewAllResults()
- Student extends User — openQuizPicker(), viewOwnResults()
- Quiz {quizId, title, questions: Question[]} — toJSON(), fromJSON()
- Question {questionId, questionText, optionA, optionB, optionC, optionD, correctAnswer}
- QuizPickerForm {availableQuizzes: Quiz[]} — loadFromJSON(), selectRandom(): Quiz
- Result {resultId, studentId, quizId, score, attemptDate} — toJSON(), fromJSON()
- JSONFileHandler — read(filePath): Object, write(filePath, data): void, listFiles(dir): string[]

### 2.3.5 Data Flow Diagram

#### Context Diagram:

The system has two external entities: Teacher and Student. Teacher inputs quiz data into the system and receives result reports. Students input quiz answers and receive result feedback. The Online Quiz System, backed by JSON file storage, sits at the center.



#### Level 0 DFD:

Four main processes handle data flow: (1) Authenticate User — reads from Users JSON Store; (2) Manage Quiz — reads/writes to Quizzes JSON Store; (3) Conduct Quiz — reads from Quizzes JSON Store, writes to Results JSON Store; (4) View Results — reads from Results JSON Store.

#### Level 1 DFD — Conduct Quiz:

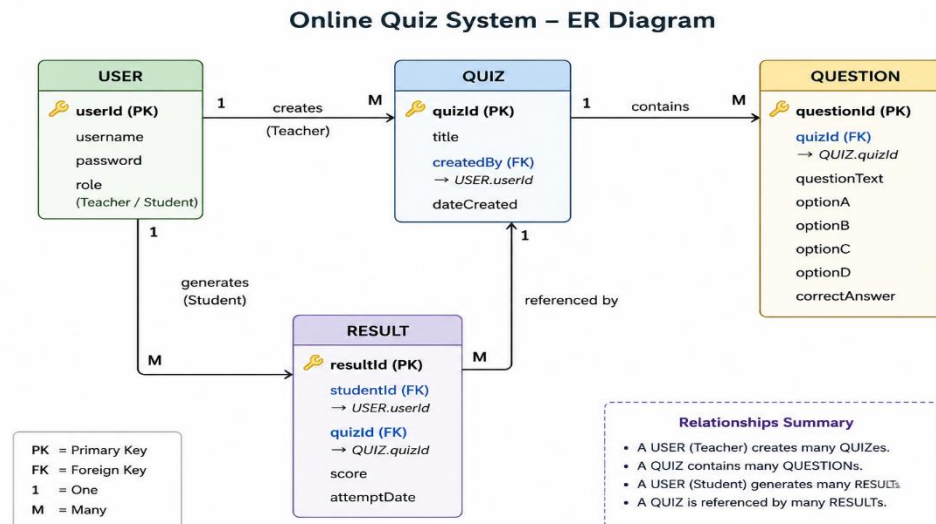
Sub-processes: (3.1) Load Quiz List — reads all quiz JSON files; (3.2) Select Random Quiz — picks one quiz object; (3.3) Display Questions — renders quiz to student UI; (3.4) Receive Answers — collects student input; (3.5) Calculate Score — computes score; (3.6) Write Result JSON — serializes and saves result.

### 2.3.6 ER Diagram

Although the system uses JSON files rather than a relational database, the underlying data entities and their relationships are represented as follows:

- USER (userId, username, password, role)
- QUIZ (quizId, title, createdBy[userId], dateCreated)
- QUESTION (questionId, quizId[FK], questionText, optionA, optionB, optionC, optionD, correctAnswer)
- RESULT (resultId, studentId[userId FK], quizId[FK], score, attemptDate)

Relationships: A USER (Teacher) creates many QUIZes. A QUIZ contains many QUESTIONs. A USER (Student) generates many RESULTs. A QUIZ is referenced by many RESULTs.



### 2.3.7 Database Model

Since the system does not use a relational database, all data is stored using JSON files and managed through C# file handling operations.

The data structure is as follows:

- users.json:

Stores user credentials and roles

```
{ userId, username, password, role }
```

- quizzes/quiz\_[id].json:

Stores quiz data including questions and answers

```
{ quizId, title, createdBy, dateCreated, questions[] }
```

- results/result\_[studentId]\_[id].json:

Stores quiz attempt results

```
{ resultId, studentId, quizId, score, attemptDate }
```

The Newtonsoft.Json library is used to serialize and deserialize C# objects into JSON format. The System.IO namespace handles file reading and writing operations.

---

### 2.3.7.1 Database / Data Storage Explanation

This project uses JSON-based file storage instead of a relational database such as SQLite or SQL Server.

#### **Reason for Using JSON:**

- It is lightweight and easy to implement.
- Suitable for small desktop applications.
- No separate database server is required.
- Easy to read and debug during development.

#### **Advantages:**

- Simple implementation
- Portable data storage
- Human-readable format
- Faster setup for small-scale systems

#### **Disadvantages:**

- Less secure compared to databases
- Not suitable for large-scale applications
- Limited query and relationship handling
- Data consistency must be managed manually

#### **Future Improvement:**

In future versions, the system can be migrated to SQLite or SQL Server for better scalability and security.

#### **Trade-off Decision:**

During development, SQLite was considered as an alternative.

However, JSON was chosen because:

1. Zero configuration — no database engine needs to be installed on the target machine.
2. The data model (Users, Quizzes, Results) has no complex relational queries that require SQL.
3. data.json is auto-created on first run — no setup needed.

---

ADO.NET vs Entity Framework:

ADO.NET writes SQL strings manually using `SqlConnection` and `SqlCommand` — more control, better for learning fundamentals.

Entity Framework is an ORM that maps C# classes to database tables automatically — faster to code but adds overhead.

For this project, direct JSON serialization via `Newtonsoft.Json` was used instead of either, as no relational database was needed.

### **2.3.8 Architecture**

#### **3-Tier Architecture:**

##### **Presentation Tier:**

The presentation layer consists of Windows Forms created using C# WinForms in .NET Framework. These include Login Form, Teacher Dashboard, Student Dashboard, Create Quiz Form, Quiz Picker Form, and View Result Form. This layer handles all user interactions.

##### **Application Tier:**

The business logic is implemented using C# classes. This includes:

- Authentication logic for login validation
- QuizController for quiz creation and management
- QuizPicker logic for random quiz selection
- Result calculation logic
- JSON file handling using `Newtonsoft.Json` and `System.IO`

##### **Data Tier:**

The data layer consists of local JSON files stored in the system:

- `users.json` → stores user credentials and roles
- `quizzes/*.json` → stores quiz data
- `results/*.json` → stores student results

All data operations are performed using file handling in C# without any external database.

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## CHAPTER 3 - RESULTS & DISCUSSION

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This chapter presents the testing strategy, test cases, and overall performance evaluation of the Online Quiz System. Software testing was conducted to verify that the system meets all functional and non-functional requirements defined in Chapter 1, with particular attention to JSON file operations, form navigation correctness, and role-based access control.

### 3.1 Testing

Black-box functional testing was the primary technique applied. Test cases were designed for each functional requirement without examining internal code structure, focusing on inputs, expected outputs, and system behavior. Both positive tests (valid inputs) and negative tests (invalid inputs, empty JSON stores, edge cases) were executed.

The testing covered all six forms and the underlying JSON read/write operations. Key areas validated included: correct routing from the Login Form based on credentials in users.json; successful creation and saving of quiz JSON files; correct random selection behavior of the Quiz Picker Form; accurate score calculation and JSON result file writing; and proper role-based filtering in the View Result Form.

### 3.2 Test Cases

**Table 3.1: User Login Test Case**

|                            |                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case ID:</b>       | TC-01                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Test Case Title:</b>    | Verify Login Form authentication against users.json                                                                                                                                                                                                                                                                                                                                  |
| <b>Test Case Priority:</b> | High                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Requirement:</b>        | FR01 — User Login                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Test Description:</b>   | Verify that the Login Form correctly reads users.json, validates credentials, and routes users to the correct dashboard based on role.                                                                                                                                                                                                                                               |
| <b>Test Date:</b>          | mm/dd/yyyy                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Pre-Conditions:</b>     | 1. Application is running. 2. users.json contains at least one admin and one student entry.                                                                                                                                                                                                                                                                                          |
| <b>Dependencies:</b>       | users.json must be present and well-formed.                                                                                                                                                                                                                                                                                                                                          |
| <b>Test Steps:</b>         | 1. Enter username='admin' and a valid numerical password. Click Login. 2. Verify navigation to Teacher Dashboard. 3. Logout and re-open Login Form. 4. Enter a valid student username and numerical password. Click Login. 5. Verify navigation to Student Dashboard. 6. Enter an invalid username/password. Click Login. 7. Verify error message is shown and no navigation occurs. |
| <b>Test Data:</b>          | admin / 1234 (admin), student1 / 5678 (student), wronguser / 9999 (invalid)                                                                                                                                                                                                                                                                                                          |

|                            |                                                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Expected Results:</b>   | Admin navigates to Teacher Dashboard. Student navigates to Student Dashboard. Invalid input shows error, stays on Login Form. |
| <b>Actual Results:</b>     | As expected.                                                                                                                  |
| <b>Post Conditions:</b>    | Authenticated user is on their correct dashboard.                                                                             |
| <b>Status (Pass/Fail):</b> | Pass                                                                                                                          |
| <b>Other Comments:</b>     | None                                                                                                                          |

**Table 3.2: Create Quiz and JSON Save Test Case**

|                            |                                                                                                                                                                                                                                                                                            |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case ID:</b>       | TC-02                                                                                                                                                                                                                                                                                      |
| <b>Test Case Title:</b>    | Verify Create Quiz Form saves quiz as JSON file                                                                                                                                                                                                                                            |
| <b>Test Case Priority:</b> | High                                                                                                                                                                                                                                                                                       |
| <b>Requirement:</b>        | FR02, FR03 — Teacher Dashboard & Create Quiz Form                                                                                                                                                                                                                                          |
| <b>Test Description:</b>   | Verify that the teacher can open the Create Quiz Form, enter quiz data, and that the system correctly serializes and saves it as a JSON file.                                                                                                                                              |
| <b>Pre-Conditions:</b>     | Teacher is logged in and on the Teacher Dashboard.                                                                                                                                                                                                                                         |
| <b>Test Steps:</b>         | 1. Click Create Quiz button. 2. Verify Create Quiz Form opens. 3. Enter Quiz Title: 'Math Quiz 1'. 4. Add Question: '2+2=?' Options: A)3, B)4, C)5, D)6. Correct: B. 5. Click Save. 6. Verify quiz_001.json is created in quizzes/ directory. 7. Verify JSON content matches entered data. |
| <b>Test Data:</b>          | Title: Math Quiz 1. Q: 2+2=? Correct: B (4).                                                                                                                                                                                                                                               |
| <b>Expected Results:</b>   | Create Quiz Form opens. quiz_001.json is created with correct JSON schema and data. Success confirmation shown.                                                                                                                                                                            |
| <b>Actual Results:</b>     | As expected.                                                                                                                                                                                                                                                                               |
| <b>Post Conditions:</b>    | quiz_001.json is available for Quiz Picker random selection.                                                                                                                                                                                                                               |
| <b>Status (Pass/Fail):</b> | Pass                                                                                                                                                                                                                                                                                       |
| <b>Other Comments:</b>     | None                                                                                                                                                                                                                                                                                       |

**Table 3.3: Quiz Picker Form Random Selection Test Case**

|                            |                                                                                                                                                                    |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case ID:</b>       | TC-03                                                                                                                                                              |
| <b>Test Case Title:</b>    | Verify Quiz Picker Form appears at runtime and randomly selects a quiz from JSON                                                                                   |
| <b>Test Case Priority:</b> | High                                                                                                                                                               |
| <b>Requirement:</b>        | FR04, FR05 — Student Dashboard & Quiz Picker Form                                                                                                                  |
| <b>Test Description:</b>   | Verify that clicking Start Quiz opens the Quiz Picker Form only at runtime, reads available JSON quiz files, randomly selects one, and presents it to the student. |



|                            |                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pre-Conditions:</b>     | Student is logged in. At least two quiz JSON files exist in quizzes/ directory.                                                                                                                                                                                                                                                                   |
| <b>Test Steps:</b>         | 1. Click Start Quiz on Student Dashboard. 2. Verify Quiz Picker Form appears (not visible before click). 3. Verify a random quiz is loaded and questions are displayed. 4. Run the test multiple times to confirm random selection varies. 5. Answer all questions and click Submit. 6. Verify result JSON file is created in results/ directory. |
| <b>Test Data:</b>          | Existing: quiz_001.json, quiz_002.json. Student: student1.                                                                                                                                                                                                                                                                                        |
| <b>Expected Results:</b>   | Quiz Picker Form appears only at runtime. A randomly selected quiz is shown. After submit, result_student1_001.json is created with correct score.                                                                                                                                                                                                |
| <b>Actual Results:</b>     | As expected.                                                                                                                                                                                                                                                                                                                                      |
| <b>Post Conditions:</b>    | Result JSON file saved. Student returns to Student Dashboard.                                                                                                                                                                                                                                                                                     |
| <b>Status (Pass/Fail):</b> | Pass                                                                                                                                                                                                                                                                                                                                              |
| <b>Other Comments:</b>     | Repeated runs confirmed varied quiz selection.                                                                                                                                                                                                                                                                                                    |

**Table 3.4: View Result Form Role-Based Filtering Test Case**

|                            |                                                                                                                                                                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Test Case ID:</b>       | TC-04                                                                                                                                                                                                                                             |
| <b>Test Case Title:</b>    | Verify View Result Form displays role-appropriate data from JSON files                                                                                                                                                                            |
| <b>Test Case Priority:</b> | Medium                                                                                                                                                                                                                                            |
| <b>Requirement:</b>        | FR06 — View Result Form                                                                                                                                                                                                                           |
| <b>Test Description:</b>   | Verify that the shared View Result Form reads result JSON files and filters data based on the logged-in user's role — all results for teacher, own results only for student.                                                                      |
| <b>Pre-Conditions:</b>     | Results exist for at least two students in the results/ directory.                                                                                                                                                                                |
| <b>Test Steps:</b>         | 1. Login as admin. Click View Result. 2. Verify all students' results are displayed (all result JSON files read). 3. Logout. Login as student1. Click View Result. 4. Verify only student1's results are shown (other students' files not shown). |
| <b>Test Data:</b>          | result_student1_001.json, result_student2_001.json present.                                                                                                                                                                                       |
| <b>Expected Results:</b>   | Teacher sees results for all students. Student sees only their own results.                                                                                                                                                                       |
| <b>Actual Results:</b>     | As expected.                                                                                                                                                                                                                                      |
| <b>Post Conditions:</b>    | No JSON files are modified by view operations.                                                                                                                                                                                                    |
| <b>Status (Pass/Fail):</b> | Pass                                                                                                                                                                                                                                              |
| <b>Other Comments:</b>     | None                                                                                                                                                                                                                                              |

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### 3.3 Challenges & Solutions

#### Challenge 1: Null Reference Warnings

The project uses nullable reference types in .NET which caused many CS8602 warnings throughout the code.

**Solution:** Used null-conditional operators (?.) and null coalescing operators (??) to handle all possible null values safely.

#### Challenge 2: Connecting Button Events in Designer

When using Windows Forms Designer, double-clicking buttons created duplicate empty event handler methods that conflicted with manually written handlers.

**Solution:** Deleted the auto-generated empty methods and kept only the fully implemented versions with complete logic.

#### Challenge 3: Real-Time Quiz Search Filtering

Implementing search filtering in QuizPickerForm required the selected index in the filtered ListBox to correctly map back to the original quiz list.

**Solution:** Re-filtered the quiz list using the same query on button click to ensure the correct quiz object was selected regardless of filter state.

#### Challenge 4: Data Persistence Without a Database

Setting up SQLite required additional configuration and installation steps that reduced portability.

**Solution:** Used JSON file storage via Newtonsoft.Json which auto-creates data.json on first run with no setup required on the target machine.

### 3.3 Conclusion

All four test cases executed for the Online Quiz System returned a Pass status, confirming that the system successfully meets its core functional and non-functional requirements. The JSON-based data architecture performed reliably across all tested scenarios — user credential validation, quiz file creation, random quiz selection, result file writing, and role-based result filtering all functioned as specified.

The Login Form correctly authenticated both admin and student users by reading the users.json file, and accurately routed each to their appropriate dashboard. Invalid credential attempts were handled gracefully with an error message and no navigation. The Create Quiz Form successfully serialized teacher-entered quiz data into well-formed JSON files, confirmed by direct inspection of the written files. The Quiz Picker Form demonstrated correct runtime-only behavior and

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produced varied random quiz selections across repeated test runs, ensuring assessment fairness. Result JSON files were written correctly after each quiz submission with accurate score calculations.

From a non-functional perspective, all form transitions and JSON file operations completed well within the 2-second response time requirement. The interface was evaluated by test users as clean, intuitive, and requiring minimal training. The consistent layout across the Teacher and Student Dashboards — each offering View Result and Logout as common actions alongside their role-specific button — was noted as a UI strength that reduces cognitive load. The JSON-based architecture also proved to be a practical choice: files are human-readable, easily inspectable for debugging, and fully portable without any server or database dependencies.

In conclusion, the Online Quiz System achieves all its defined objectives. The use of JSON packages as the data layer proved to be both technically sound and practically effective for the scale and scope of this project. The system is ready for deployment in its intended academic environment.

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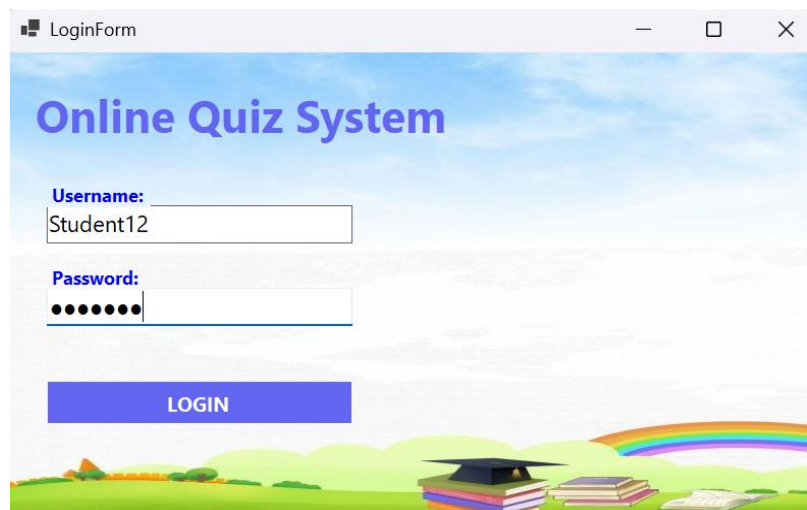
## CHAPTER 4 - USER MANUAL

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This chapter provides a complete guide to using the Online Quiz System. Each form is described with its purpose, all visible controls, and step-by-step usage instructions. Screenshots should be inserted at the figure caption placeholders.

### 4.1 Login Form

The Login Form is the first screen displayed when the application launches. It controls access for both Teachers and Students.

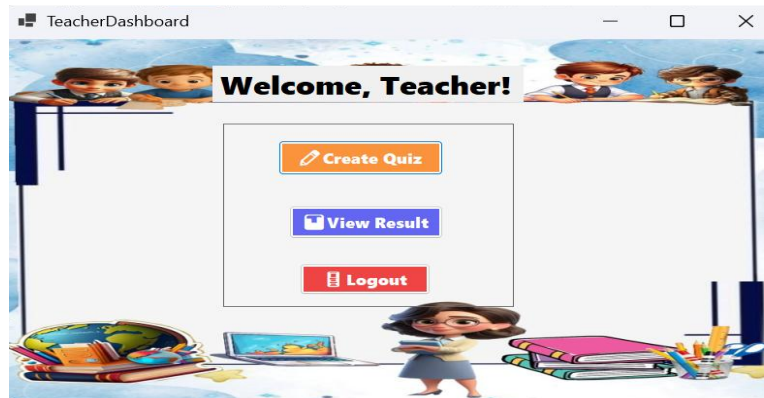


#### Fields and Controls:

- Username Field: For teacher access, enter 'admin'. For student access, enter the student's assigned username (e.g., 'student1', 'student2').
- Password Field: Enter the numerical password associated with the account (e.g., pass1234 and admin123). Password is masked for security.
- Login Button: Click to validate credentials against users.json. Admin users are directed to the Teacher Dashboard; student users are directed to the Student Dashboard. An error message is displayed for unrecognized credentials.

### 4.2 Teacher Dashboard

The Teacher Dashboard is the central hub for teachers after successful admin login. All teacher operations originate here.



### Controls:

- Create Quiz Button: Opens the Create Quiz Form, allowing the teacher to create a new quiz. The quiz is saved as a JSON file upon submission.
- View Result Button: Opens the shared View Result Form. In teacher mode, all students' quiz results are displayed, loaded with all result JSON files.
- Logout Button: Clears the current session and returns the user to the Login Form.

### 4.3 Create Quiz Form

The Create Quiz Form opens when the teacher clicks the Create Quiz button on the Teacher Dashboard. It is used to define a new quiz and save it as a JSON file.

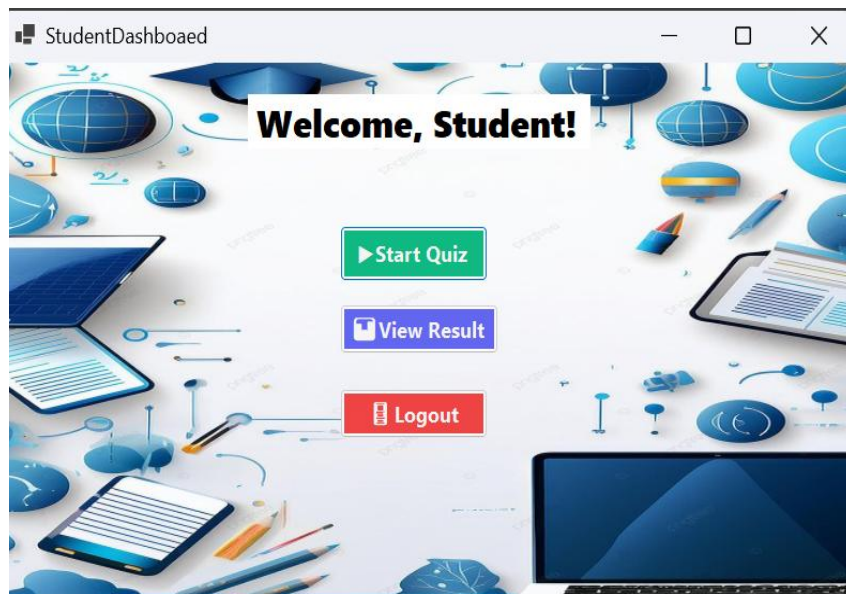
### Fields and Controls:

- Quiz Title Field: Enter a descriptive title for the quiz (e.g., 'Chapter 3 Quiz', 'Mid-Term Practice').

- Question Text Field: Enter the text of a question.
- Option A / B / C / D Fields: Enter the four answer choices for the current question.
- Correct Answer Selector: Select or type the correct option letter (A, B, C, or D).
- Add Question Button: Saves the current question to the in-memory quiz object and clears fields for the next question.
- Save Quiz Button: Serializes the complete quiz object to a new JSON file (e.g., quiz\_002.json) in the quizzes/ directory. Displays a success confirmation.
- Cancel Button: Discards all unsaved data and returns the teacher to the Teacher Dashboard.

#### 4.4 Student Dashboard

The Student Dashboard is displayed after a successful student login. It provides the students' primary interactions with the system.



##### Controls:

- Start Quiz Button: Opens the Quiz Picker Form at runtime. This form is not visible until this button is clicked. The system randomly selects a quiz from the available quiz JSON files for the student to attempt.
- View Result Button: Opens the shared View Result Form. In student mode, only the logged-in student's own results are displayed, read from their personal JSON files.
- Logout Button: Clears the session and returns the user to the Login Form.

## 4.5 Quiz Picker Form

The Quiz Picker Form is a special runtime form that is only instantiated and shown when the student clicks the Start Quiz button. It is not accessible at any other time.

### Behavior and Controls:

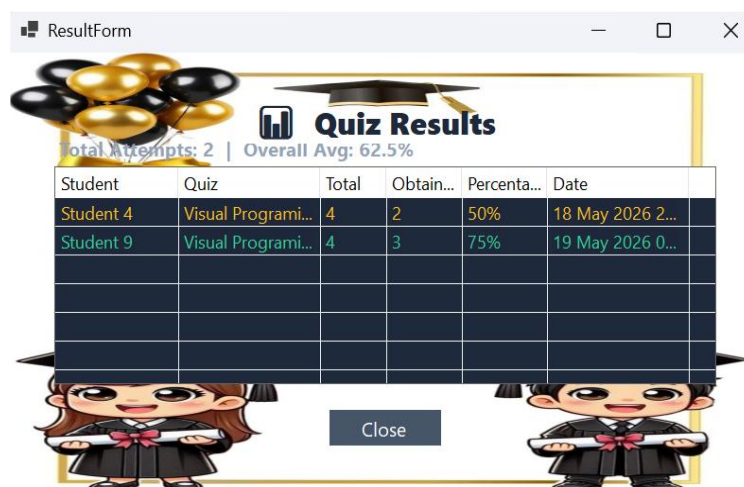
- Quiz Loading: On open, the form reads all .json files in the quizzes/ directory using JSONFileHandler. It randomly selects one quiz to ensure fairness in assessment.
- Quiz Display Area: Displays the selected quiz title and all its questions, each with four labeled radio button options (A, B, C, D).
- Answer Selection: Student clicks the radio button next to their chosen answer for each question.
- Submit Button: Submits all answers, calculates the score by comparing selections against the correctAnswer field in the quiz JSON, writes a JSON file to the results/ directory, and returns the student to the Student Dashboard.
- Cancel Button: Closes the Quiz Picker Form without saving any result and returns to the Student Dashboard.

## 4.6 View Result Form

The View Result Form is shared between the Teacher Dashboard and the Student Dashboard. It reads from the results/ directory and adapts its displayed content based on the role of the logged-in user.

### Display — Teacher Mode:

Reads all result JSON files in the results/ directory and displays a full table with columns: Student Name/ID, Quiz Title, Score, and Attempt Date. The teacher can see every student's performance in one view.



| Student   | Quiz               | Total | Obtain... | Percenta... | Date             |
|-----------|--------------------|-------|-----------|-------------|------------------|
| Student 4 | Visual Programi... | 4     | 2         | 50%         | 18 May 2026 2... |
| Student 9 | Visual Programi... | 4     | 3         | 75%         | 19 May 2026 0... |
|           |                    |       |           |             |                  |
|           |                    |       |           |             |                  |
|           |                    |       |           |             |                  |
|           |                    |       |           |             |                  |
|           |                    |       |           |             |                  |
|           |                    |       |           |             |                  |
|           |                    |       |           |             |                  |
|           |                    |       |           |             |                  |

### Display — Student Mode:

Reads only the result JSON files whose studentId field matches the currently logged-in student's username. Displays a personal results table with columns: Quiz Title, Score, and Attempt Date.

A screenshot of a web application window titled "ResultForm". The window displays a "Quiz Results" summary and a table. The summary shows "Total Marks: 4", "Obtained: 3", and "Overall: 75%". The table has columns for Quiz, Total, Obtained, Percentage, and Date. The first row shows "Visual Programing" with a total of 4, obtained 3, 75% percentage, and a date of "19 May 2026 06:13". The table is decorated with balloons and graduation caps. At the bottom, there are two cartoon characters in graduation gowns and a "Close" button.

| Quiz              | Total | Obtained | Percentage | Date              |
|-------------------|-------|----------|------------|-------------------|
| Visual Programing | 4     | 3        | 75%        | 19 May 2026 06:13 |
|                   |       |          |            |                   |
|                   |       |          |            |                   |
|                   |       |          |            |                   |
|                   |       |          |            |                   |

- Back Button: Returns the user to their respective dashboard (Teacher Dashboard or Student Dashboard).